

Average	1.8	1.2	1.25	-	2	-	-	-	2	-	-	-	2	-	1.33
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1 – Weakly Mapped (Low)

2 – Moderately Mapped (Medium)

3 – Strongly Mapped (High)

“ - ” means there is no correlation

Syllabus

Unit I:

12 Lecture Hours

Introduction to Computer Graphics and OpenGL: Overview of Computer Graphics, Computer Graphics Application and Software, Description of some graphics devices, Introduction to pixel. Why OpenGL, features in OpenGL, OpenGL operations, Abstractions in OpenGL – GL, GLU & GLUT, 3D viewing pipeline, viewing matrix specifications, a few examples and demos of OpenGL programs.

Unit II:

12 Lecture Hours

Scan conversion – lines, circles and Ellipses; Filling polygons and clipping algorithms Scan Converting Lines, Mid-point criteria, Aliasing and Antialiasing, Problems of Aliasing, end-point ordering and clipping lines, Scan Converting Circles, Scan Converting Ellipses, Filling Polygons Clipping Lines algorithms– Cyrus Beck, Cohen-Sutherland and Liang-Barsky, Clipping Polygons, problem with multiple components.

Unit III:

12 Lecture Hours

2-D and 3-D Transformations: Transformations and Matrices, Transformation Conventions, 2D Transformations, Homogeneous Coordinates and Matrix Representation of 2D Transformations, Translations and Homogeneous Coordinates, Rotation, Reflection, Scaling, Combined Transformation, Transformation of Points, Transformation of The UNIT Square, Rotation About an Arbitrary Point, Reflection through an Arbitrary Line, A Geometric Interpretation of Homogeneous Coordinates, The Window-to-Viewport Transformations. Three-Dimensional Scaling, Three-Dimensional Shearing, Three-Dimensional Rotation, Three Dimensional Reflection, Three-Dimensional Translation, Multiple Transformation, Rotation about an Arbitrary

Axis in Space, Reflection through an Arbitrary Plane, Matrix Representation of 3D Transformations, Composition of 3D Transformations.

Unit IV:

12 Lecture Hours

Rendering : Visible-Surface Determination, Techniques for efficient Visible-Surface Algorithms, Categories of algorithms, Back face removal, the Z-Buffer Algorithm, Scan-line method, Painter's algorithms (depth sorting), Octrees , BSP trees, Visible-Surface Ray Tracing, comparison of the methods.

Illumination and Shading Models for Polygons, Reflectance properties of surfaces, Ambient, Specular and Diffuse reflections, Phong's model, Gouraud shading, some examples.

Unit V:

12 Lecture Hours

Plane Curves and Surfaces

Curve Representation, Nonparametric Curves, Cubic Splines, Bezier Curves, Bspline Curves, B-spline Curve Fit, Knot Vectors, NURBS, Parametric Cubic Curves, Quadric Surfaces. Bezier Surfaces, Fractals.

Total lecture Hours 60

Textbooks

1. Donald D. Hearn and M. Pauline Baker, "Computer Graphics using OpenGL", Pearson Education India, 2014.

Reference Books

1. David F. Rogers and J. Alan Adams, "Mathematical Elements for Computer Graphics", 2nd Edition, McGraw- Hill, Inc, 2017.
2. James D. Foley, Andries van Dam, Steven K. Feiner, and John F. Hughes "Computer Graphics: Principles and Practice", Pearson, 2002.

Modes of Evaluation: Quiz/Assignment/ presentation/ extempore/ Written Examination